Venkata Lakshmi Parimala Pasupuleti

Senior AI Developer · Multifamily Housing & Property Management · Agent Design · Evals · Guardrails · Azure OpenAI · Python · Production AI

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SUMMARY

Software engineer with **7+ years of experience**, including deep expertise in **multifamily housing and property management workflows** - and now building production AI that fits how that work actually moves. Spent 7+ years at **Entrata** (affordable housing SaaS, 100K+ monthly users): owned HUD compliance ETL (180K records), async rent processing across thousands of tenant records, PHI-compliant resident data, and multi-step property operations workflows - I know these processes from the inside. Currently at Intuit: **AI agents in production** with tool calling, structured outputs, HITL guardrails, retrieval-grounded context, and CI eval harnesses using Anthropic Claude. Built **Leasel** - a production **lease compliance AI platform**: 8-agent LangGraph orchestration, document RAG (OCR ingestion, hybrid search, citation grounding), HITL guardrails for regulated decisions, MCP server integration, and a CI eval harness (93.6% precision, 1,240 test cases). Built **AgentProbe** - an open-source eval and observability platform with **Azure OpenAI** routing, CI regression gating, and per-run tracing. Python, FastAPI, Azure OpenAI, AWS, SQL, LangGraph. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

Multifamily Housing · Property Management · Affordable Housing · HUD · LIHTC

Entrata · Yardi · RealPage · Property Management Platforms

Agent Design · Skill Orchestration · Tool Calling · Multi-Step Workflows

Guardrails · HITL · Human Approvals · Safety Controls · Workflow Boundaries

Eval Frameworks · Agent Task Success · Quality Metrics · CI Regression

RAG · Document Understanding · OCR · Retrieval · Citation Grounding

Azure OpenAI · Azure AI Foundry · AWS · Cloud Deployment · Observability

Python · FastAPI · SQL · LangGraph · LangChain · MCP

Workflow Automation · Business Process AI · Operational AI

Structured Outputs · Prompt Engineering · Model Inference · Cost Management

Production Reliability · Logging · Monitoring · Rollback · CI/CD

PHI-Compliant Systems · IAM · Least Privilege · Secure AI Design

EXPERIENCE

Software Engineer

Jun 2026 - Present

Intuit — Virginia, USA (Remote)

Financial services AI - LLM-powered agents and workflows in production

- **Built production AI agents** with Anthropic Claude: multi-step tool calling, structured outputs, retrieval-grounded context, HITL human approvals for sensitive decisions, and retries with exponential backoff - same pattern required for property management workflow automation.
- **Designed and owns the LLM eval harness:** CI regression gating with golden datasets and LLM-as-judge scoring; structured logging per agent run; qualitative failure analysis converted to eval coverage - ensuring agents behave reliably in production before and after every release.
- Build ML data pipelines (scikit-learn, XGBoost); React/TypeScript front ends; deploy via Docker + GitHub Actions CI/CD to AWS with CloudWatch monitoring and observability.

Senior Software Engineer

Oct 2020 - Jul 2024

Entrata — Affordable Housing & Multifamily SaaS

Regulated multi-tenant property management platform · 100K+ monthly active users · Affordable housing, market-rate multifamily, LIHTC, HUD

- **HUD compliance workflows:** Owned the annual HUD income-limit import for 6 consecutive years - loading federal housing data into 180,000 records with validation, rollback checks, and zero compliance violations. Deep familiarity with affordable housing regulatory requirements and data quality standards.
- **Async property operations processing:** Built RabbitMQ-based event-driven system handling rent updates across thousands of tenant records - idempotency guarantees, retry logic, dead-letter routing, and PHI-compliant resident data handling (Libsodium encryption, IAM least-privilege, AWS S3 migration, 100% coverage, zero incidents).
- Led Docker microservices redesign (50% scalability improvement); optimized 200+ SQL queries (30% performance gain); Python/Bash automation eliminating 80% of manual operational work; 300+ code reviews; mentored 5+ engineers across 120+ Agile sprints.

Junior Software Engineer

Sep 2018 - Oct 2020

Entrata India — Pune

Property management platform - affordable housing and market-rate multifamily modules

- Built property management modules (Python, PHP, Java, PostgreSQL); delivered large-scale housing data integration workflows (1.6M HUD/Tax Credit records); created 56 state-specific housing certification documents; led AWS S3 migration from legacy storage.

PROJECTS

Leasel - Lease Compliance AI Platform: Agent Design · RAG · Guardrails · Evals

2026 | Python, LangGraph, FastAPI, ChromaDB, MCP, CI/CD | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Production **lease compliance AI platform** - directly applicable to property management operations: **8-agent LangGraph orchestration** (triage, retrieval, reasoning, validation) with tool calling, skill design, and conditional routing; **document RAG** (OCR ingestion, section-aware chunking, hybrid TF-IDF + semantic reranking, citation-grounded answers); **HITL guardrails** so eligibility and financial decisions always route to staff; **MCP server integration** for shared tool access; **CI eval harness** (LLM-as-judge, 93.6% precision, 1,240 test cases). Iterated from prototype to 3 production surfaces with full observability.

AgentProbe - Agent Eval & Observability Platform with Azure OpenAI Routing

2025 | Python, FastAPI, PostgreSQL, Azure OpenAI, CI/CD | github.com/venkatalakshmiparimala/AgentProbe

- Open-source platform for **AI agent quality and reliability**: offline golden-dataset evals, LLM-as-judge scoring for agent task success, CI regression gating, structured failure classification; cost-aware model routing including **Azure OpenAI**, Anthropic, OpenAI, AWS Bedrock, Groq, and Ollama; full per-run tracing, logging, and instrumentation. Used as a production release gate before every AI deployment.

AI on FHIR — Workflow Automation via Document & API Integration

2024 | Python, Flask, React, BERT, LLMs, FHIR REST APIs, Docker | github.com/venkatalakshmiparimala/AI_on_FHIR

- Built an AI system that automates multi-step document and API workflows: translates natural-language questions into FHIR REST API calls using LLMs, BERT, and embeddings - cut lookup time ~60%. Demonstrates the core pattern of **connecting AI to regulated operational data**: OAuth-secured API integration, HIPAA-style RBAC, least-privilege access, Docker-containerized deployment.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia | GPA 3.87 May 2026

B.Tech Computer Science & Engineering — JNTU Anantapur 2017